

OpsFlow 代码结构

代码结构 — 目录树 + 模型关系 + API 端点 + 模块依赖

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1. 后端目录树 (backend/opsflow/)

```
opsflow/
├── __init__.py
├── apps.py                # Django AppConfig (ready: 自动发现插件)
├── consumers.py           # WebSocket consumers (执行状态推送)
├── tasks.py               # Celery 任务定义
├── urls.py                # URL 路由注册
├── serializers.py         # DRF Serializers
├── schema/               # Pydantic Schema (form_schema)
│   └── form_schema.py     # FormConfig: 表单配置定义
├──
├── models/               # 数据模型 (拆分模块)
│   ├── __init__.py       # 统一重新导出
│   ├── template.py        # FlowTemplate, TemplateVersion, TemplateNode, TemplateCollect, TemplateCategory
│   ├── execution.py       # FlowExecution, ExecutionNode, ExecutionScheme, AutoRetryStrategy, NodeTimeoutConfig, NodeExecutionTrace
│   ├── project.py         # OpsProject, ProjectMember
│   ├── plugin.py          # PluginMeta
│   ├── schedule.py        # SchedulePlan
│   ├── webhook.py         # WebhookConfig, WebhookLog
│   ├── audit.py           # OpsLog, OperationRecord
│   ├── knowledge.py       # OpsKnowledge
│   ├── auth.py            # ApiToken
│   └── env.py             # ProjectEnvironmentVariable
├──
├── core/                 # 核心引擎模块
│   ├── __init__.py
│   ├── flow_engine.py     # FlowEngine — 流程执行引擎
│   ├── states.py          # NodeState / PipelineState 枚举 + 流转矩阵
│   ├── pipeline_builder/  # Pipeline Tree 构建器
│   │   ├── __init__.py    # build_bamboo_pipeline() 主入口
│   │   ├── conditions.py  # 边条件解析
│   │   ├── elements.py    # 节点元素创建 (ServiceActivity/Gateway/SubProcess)
│   │   └── validation.py  # 循环引用检测
│   ├── plugin_service_adapter.py # PluginService + OpsflowPluginComponent
│   ├── node_dispatcher.py # NodeCommandDispatcher — 节点操作调度
│   ├── pipeline_schema.py # Pipeline Tree JSON Schema 校验
│   ├── pipeline_preview.py # Pipeline 预览 (简化 build)
│   ├── safety_guard.py    # 安全校验 (白名单/重试上限/循环引用)
│   ├── bamboo_validator.py # bamboo-engine 兼容性校验
│   ├── auto_retry.py      # 自动重试策略创建 + 派发
│   ├── node_timeout_strategy.py # Redis 分布式超时追踪 + 策略执行
│   ├── llm_service.py     # AI 流程生成 (DeepSeek API + RAG)
│   ├── mako_resolver.py   # Mako 模板引擎变量解析
│   └── variable_resolver.py # 变量 ${key} 替换 + 清理
```

```
| |—— variable_registry.py      # 系统变量注册 (pipeline_start/pipeline_end 等)
| |—— variables/                # 变量类型实现
| |   |—— __init__.py           # lazy_resolver 延迟计算
| |   |—— common.py             # 公共变量 (target_host, loop 等)
| |—— node_sync.py              # 节点持久化同步 (TemplateNode/ExecutionNode)
| |—— trace_logger.py           # 节点轨迹日志写入器 (JSON Lines)
| |—— webhook_service.py        # Webhook 回调服务 (HMAC + 重试)
| |—— subprocess_dispatcher.py  # 独立子流程调度器
| |—— scheduler_service.py      # APScheduler 定时调度服务
| |—— tower_service.py          # Ansible Tower/AWX 集成
| |—— conflict_checker.py       # 并发冲突检测
| |—— audit_logger.py           # 审计日志
| |—— error_codes.py            # 错误码定义 + api_success/api_error
| |—— plugin_deprecation.py     # 插件弃用检查
| |—— layout/                  # Sugiyama 分层布局引擎
| |   |—— __init__.py
| |   |—— constants.py         # NodeType, POSITION, CANVAS_WIDTH
| |   |—— layout_adapter.py     # opsflow pipeline 格式适配 + compute_layout 入口
| |   |—— normalize.py         # 输入标准化
| |   |—— acyclic.py           # 环边反转
| |   |—— drawing.py           # 主绘图流程
| |   |—— position.py          # 坐标分配
| |   |—— dummy.py             # 虚拟节点
| |   |—— utils.py             # 工具函数 (层次属性计算)
| |   |—— tests.py             # 布局测试
| |   |—— order/               # 交叉最小化
| |   |   |—— __init__.py
| |   |   |—— builder.py       # 层构建
| |   |   |—— order.py         # WDA 排序算法
| |   |—— rank/                # 层级分配
| |   |   |—— __init__.py
| |   |   |—— longest_path.py  # 最长路径层级
| |   |   |—— feasible_tree.py # 紧树构造
| |   |   |—— tight_tree.py    # 紧树搜索
| |   |   |—— utils.py         # 层级工具
| |—— plugins/                  # 标准插件集合
| |   |—— base.py              # BasePlugin 基类
| |   |—— registry.py          # 插件注册中心 (PLUGIN_REGISTRY)
| |   |—— ansible/             # Ansible 执行插件
| |       |—— shell.py, script_exec.py, file_copy.py
| |       |—— upload_file.py, backup_file.py
| |       |—— java_deploy.py, docker_deploy.py
```

```
| | |—— nginx_reload.py, service_control.py
| |—— common/                # 通用插件
| | |—— send_alert.py, test_print_time.py
| |—— esxi/                  # VMware ESXi 插件
| | |—— create_vm.py, destroy_vm.py, get_state.py
| | |—— power_on.py, power_off.py
| |—— http/                  # HTTP API 调用插件
| | |—— api_call.py
| |—— monitor/              # 监控插件
| | |—— disk_check.py, health_check.py, ping_test.py
| |—— netapp/               # NetApp 存储插件
| | |—— netapp_create_snapshot.py, netapp_create_volume.py
| | |—— netapp_delete_volume.py, netapp_get_volume.py
| | |—— netapp_modify_volume.py
| |—— pmax/                 # Dell PowerMax 插件
| | |—— performance.py, snapshot.py, storage_group.py
| |—— redfish/              # Redfish BMC 插件
| | |—— redfish_firmware_inventory.py, redfish_get_system_info.py
| | |—— redfish_list_storage.py, redfish_power_cycle.py
| |—— servicenow/           # ServiceNow ITSM 插件
| |—— verify/               # 验证测试插件
|
|—— signals/                # 信号处理模块
| |—— __init__.py
| |—— handlers.py           # on_post_set_state 主接收器
| |—— state.py              # 节点状态持久化 (JSON_SET 原子更新)
| |—— trace.py              # 轨迹记录 + 日志写入
| |—— notify.py             # WebSocket 推送
| |—— timeout.py            # Redis 超时追踪更新
| |—— helpers.py            # 信号辅助函数
|
|—— views/                  # API 视图
| |—— base.py               # ProjectFilteredViewSet, ProjectReadOnlyViewSet
| |—— template_views.py     # FlowTemplateViewSet (CRUD)
| |—— execution_views.py    # FlowExecutionViewSet (CRUD + dry_run)
| |—— plugin_views.py       # PluginViewSet
| |—— project_views.py      # OpsProjectViewSet
| |—— schedule_views.py     # SchedulePlanViewSet
| |—— scheme_views.py       # ExecutionSchemeViewSet
| |—— node_views.py         # 节点状态/轨迹查询
| |—— audit_views.py        # 审计日志查询
| |—— log_views.py          # 操作日志查询
| |—— knowledge_views.py    # 知识库视图
```

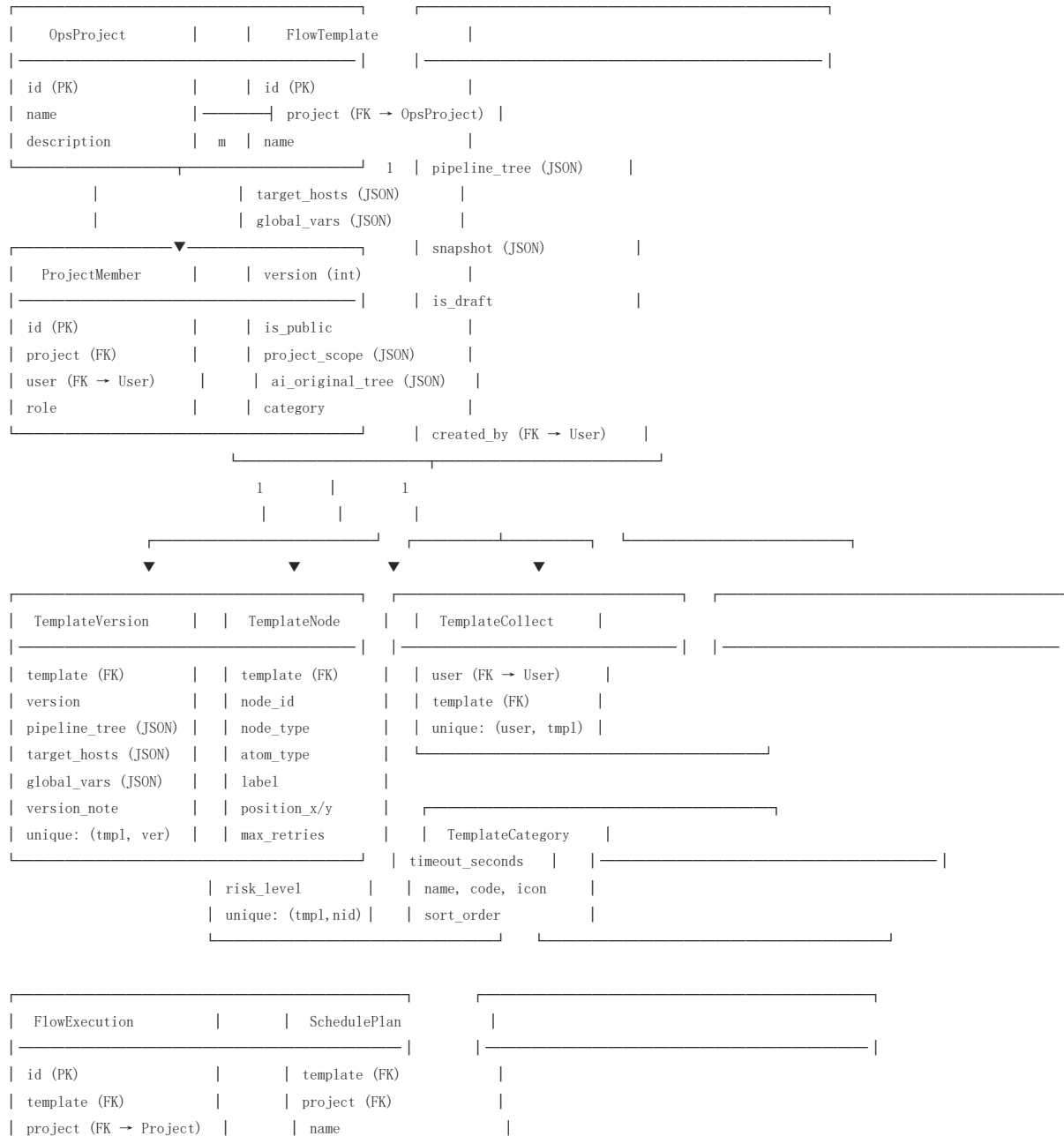
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| | └── template_category_views.py      # 模板分类视图
| | └── mixins/                        # ViewSet Mixin 拆分
| |   ├── template_ai.py              # AI 生成/分析/布局/比较
| |   ├── template_version.py         # 版本管理 (发布/回滚/diff)
| |   ├── template_variable.py        # 变量管理
| |   ├── template_subprocess.py      # 子流程追踪
| |   ├── template_export.py          # 导入导出
| |   ├── template_collect.py         # 收藏
| |   ├── template_webhook.py         # Webhook 配置
| |   ├── execution_lifecycle.py       # 生命周期 (start/pause/resume/cancel)
| |   ├── execution_node_command.py   # 节点操作 (retry/skip/force_fail/batch)
| |   ├── execution_approval.py       # 审批 (approve/reject)
| |   └── execution_trace.py          # 轨迹查询
| └── dashboard_views/                # 仪表盘统计
|   ├── stats.py                     # 基础统计 (总数/成功率/运行中)
|   ├── trends.py                    # 趋势统计
|   └── analytics.py                  # 深度分析 (Top模板/用户活跃/分布)
|
└── apigw/                            # BK API Gateway 适配
    ├── auth.py                      # JWT/BK 认证
    └── views.py                      # 网关视图
|
└── management/commands/              # Django 管理命令
    ├── start_opsflow_scheduler.py    # 启动 APScheduler
    ├── seed_sample_template.py       # 种子模板数据
    ├── add_opsflow_menu.py           # 添加菜单
    ├── seed_template_categories.py   # 种子分类数据
    ├── seed_knowledge.py             # 种子知识库数据
    ├── clean_node_trace_logs.py      # 清理日志
    ├── clean_opsflow_data.py         # 清理数据
    ├── fix_node_status_double_encode.py # 修复双编码
    └── opsflow_migrate_projects.py    # 项目迁移
|
└── migrations/                       # 数据库迁移 (0001-0008)
|
└── doc/                              # 文档
    ├── README.md
    ├── TODO.md / TODO.html
    ├── quick-start.md
    ├── link_rules.md
    ├── generate_pdf.py               # MD → PDF 生成工具
    ├── 注意事项.md
    └── design_plan/                 # 设计文档
```

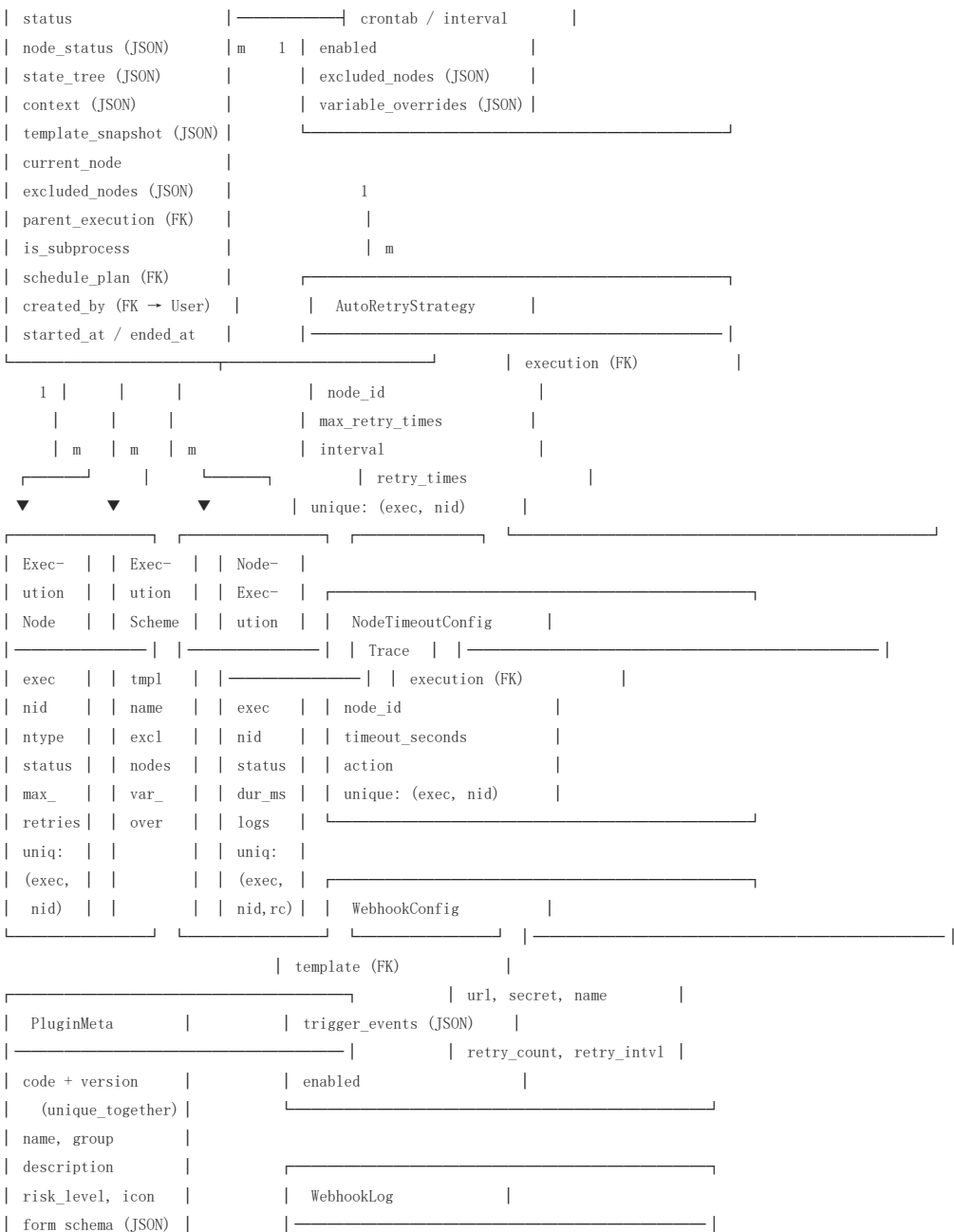
2. 前端目录树 (web/src/views/apps/opsflow/)

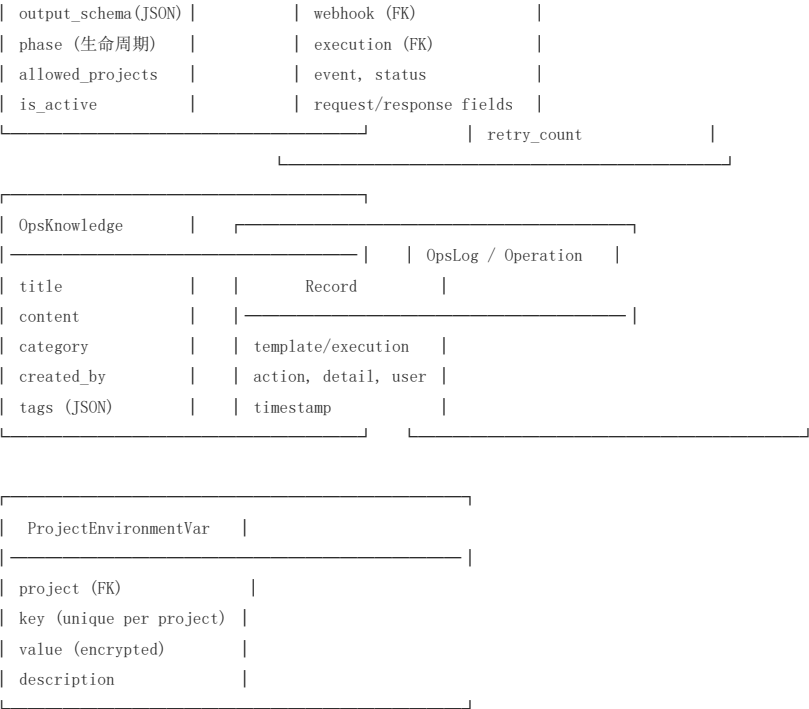
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web/src/views/apps/opsflow/
├── index.vue                # 主入口 (路由: /opsflow)
├── stores/
│   └── opsflowStore.ts      # Pinia 状态管理
├── types/
│   └── index.ts             # TypeScript 类型定义
├── utils/
│   └── shapes.ts            # X6 自定义图形 (Node/Edge 样式)
├── composables/
│   ├── useDesignCanvas.ts   # 设计器画布交互
│   ├── useMonitor.ts        # 监控画布交互
│   ├── useGraphCanvas.ts    # 通用 X6 画布操作
│   ├── useGraphValidator.ts # 画布图结构校验
│   └── useAutoSave.ts        # 模板自动保存 (debounce)
├── components/
│   ├── DesignCanvas.vue     # 流程设计器 (X6)
│   ├── MonitorCanvas.vue    # 执行监控画布 (X6)
│   ├── PropertyPanel.vue    # 节点属性面板 (右侧)
│   ├── GlobalVariablePanel.vue # 全局变量编辑
│   ├── CreateTemplateWizard.vue # 创建模板向导
│   ├── SubmitWizardDialog.vue # 提交执行向导
│   ├── PluginPickerDialog.vue # 插件选择弹窗
│   ├── PluginVisibilityDialog.vue # 插件可见性配置
│   ├── SchemeManager.vue    # 执行方案管理
│   ├── SchemeSelector.vue   # 执行方案选择
│   ├── ConditionDialog.vue  # 条件编辑弹窗
│   ├── ConditionRow.vue     # 条件行 (AND/OR)
│   ├── VariableBrowser.vue  # 变量浏览器
│   ├── VariablePicker.vue   # 变量选择器
│   ├── DiffModal.vue        # 版本 Diff 弹窗
│   ├── DryRunDialog.vue     # Dry Run 弹窗
│   ├── ProjectSwitcher.vue  # 项目切换器
│   ├── ProjectEnvVarPanel.vue # 项目环境变量
│   ├── SubprocessStatusBadge.vue # 子流程状态徽标
│   ├── HelpDrawer.vue       # 帮助抽屉
│   └── MonitorPanel.vue     # 监控面板
├── api/                     # API 请求封装
│   ├── request.ts           # axios 实例 (拦截器)
│   ├── templates.ts         # 模板相关 API
│   ├── executions.ts        # 执行相关 API
│   ├── plugins.ts           # 插件相关 API
│   ├── projects.ts          # 项目相关 API
│   └── dashboard.ts         # 仪表盘 API
```

		schedule-plans.ts	# 定时计划 API
		webhooks.ts	# Webhook API
		audit.ts	# 审计 API
		logs.ts	# 日志 API
		knowledge.ts	# 知识库 API
		servicenow.ts	# ServiceNow API
		template-categories.ts	# 分类 API
		opsflow-approval/	
		index.vue	# 审批页面
		opsflow-dashboard/	
		index.vue	# 仪表盘页面
		opsflow-execution/	
		index.vue	# 执行列表页
		components/	
		ExecutionList.vue	# 执行列表
		ExecutionDetail.vue	# 执行详情

3. 模型关系图







4. API 端点表

ViewSet/Prefix	Method	URL	功能
FlowTemplateViewSet	GET	`/api/opsflow/templates/`	模板列表（分页）
	POST	`/api/opsflow/templates/`	创建模板
	GET	`/api/opsflow/templates/{id}/`	模板详情
	PUT	`/api/opsflow/templates/{id}/`	更新模板
	DELETE	`/api/opsflow/templates/{id}/`	删除模板
	POST	`/api/opsflow/templates/create_from_ai/`	AI 生成模板

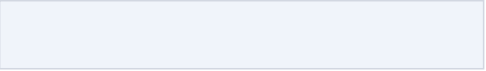


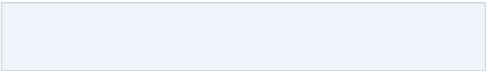
POST



``/api/opsflow/templates/analyze/``

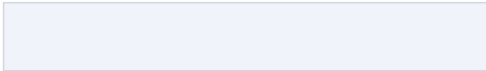






POST

``/api/opsflow/templates/refine/``



POST



``/api/opsflow/templates/ai_layout/``



自动布局

POST

``/{id}/confirm_draft/``

确认草稿 (V1)

POST

`~/id/publish/`

[发布新版本](#)

GET

`~/id/versions/`

版本列表

POST

``/{id}/rollback/``

[回滚版本](#)

POST

``/{id}/version_diff/``

版本对比

GET

`/{id}/diff\_draft/`

草稿 vs 已发布 diff

GET

``/{id}/diff/``

GET

``/{id}/subprocess_chain/``

POST

``/{id}/collect/``

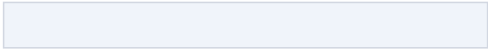
[收藏/取消收藏](#)

GET

``/{id}/webhooks/``

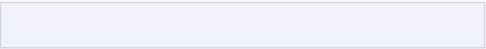
ViewSet/Prefix	Method	URL	功能
	POST	`/{id}/webhooks/`	更新 Webhook 配置
	GET	`/{id}/check_deprecated_plugins/`	弃用插件检查
	POST	`/api/opsflow/templates/update_plugin_phase/`	更新插件生命周期
FlowExecutionViewSet	GET	`/api/opsflow/executions/`	执行列表 (用户过滤)
	POST	`/api/opsflow/executions/`	创建执行
	GET	`/api/opsflow/executions/{id}/`	执行详情
	POST	`/{id}/start/`	启动执行
	POST	`/{id}/pause/`	暂停执行
	POST	`/{id}/resume/`	恢复执行
	POST	`/{id}/cancel/`	取消执行
	POST	`/{id}/retry_node/`	重试节点
	POST	`/{id}/skip_node/`	跳过节点
	POST	`/{id}/force_fail/`	强制失败
	POST	`/{id}/batch_retry/`	批量重试
	POST	`/{id}/batch_skip/`	批量跳过
	POST	`/{id}/retry_subprocess/`	重试子流程
	GET	`/{id}/trace/{node_id}/`	节点轨迹
	GET	`/{id}/trace_log/{node_id}/`	节点日志
	GET	`/{id}/state_tree/`	状态树
	GET	`/{id}/traces/`	全部轨迹摘要
	GET	`/api/opsflow/executions/pending_approval/`	待审批列表
	POST	`/{id}/approve/`	审批通过
	POST	`/{id}/reject/`	审批拒绝
	POST	`/api/opsflow/executions/dry_run/`	Dry Run 执行

****PluginViewSet****

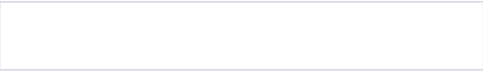


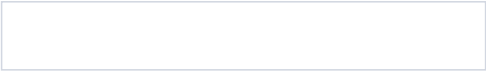
GET

`~/api/opsflow/plugins/``



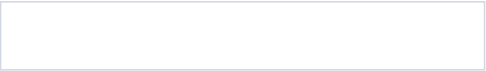
插件列表（按组分）





POST

``/api/opsflow/plugins/sync/``



同步插件元数据

POST

`~/api/opsflow/plugins/update_phases/``

批量更新生命周期

****SchedulePlanViewSet****

CRUD

`~/api/opsflow/schedule-plans/``

****OpsProjectViewSet****

CRUD

`~/api/opsflow/projects/``

****ExecutionSchemeViewSet****

CRUD

``/api/opsflow/schemes/``

****Dashboard****

GET

``/api/opsflow/dashboard/stats/``

GET

`~/api/opsflow/dashboard/trend/``

执行趋势

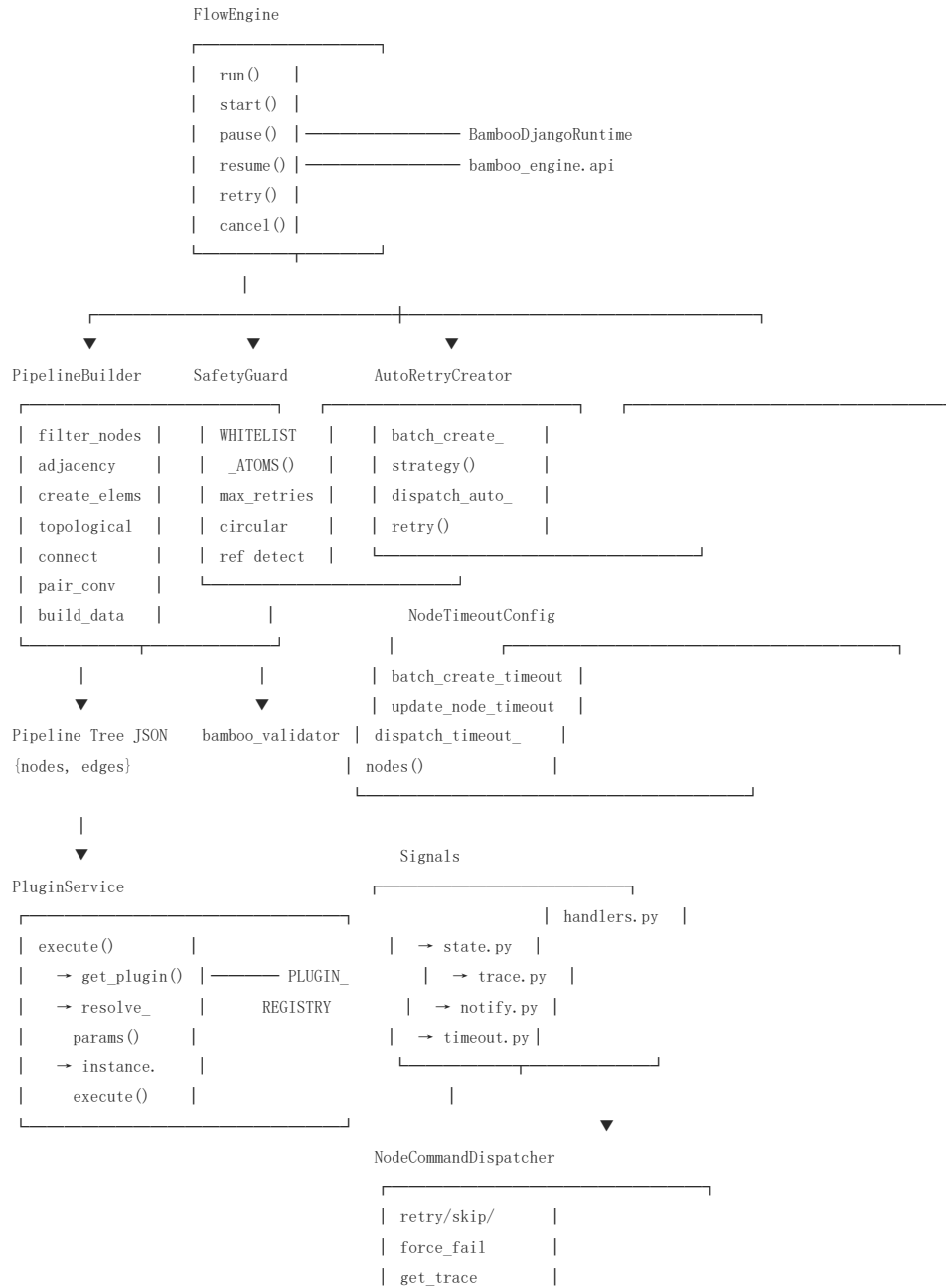
GET

`~/api/opsflow/dashboard/success_rate_trend/``

成功率趋势

ViewSet/Prefix	Method	URL	功能
	GET	`/api/opsflow/dashboard/top_templates/`	Top 模板
	GET	`/api/opsflow/dashboard/user_activity/`	用户活跃
	GET	`/api/opsflow/dashboard/status_distribution/`	状态分布
	GET	`/api/opsflow/dashboard/node_type_distribution/`	节点类型分布
	GET	`/api/opsflow/dashboard/duration_distribution/`	耗时分布
	GET	`/api/opsflow/dashboard/node_duration_top/`	节点耗时 Top
	GET	`/api/opsflow/dashboard/schedule_stats/`	定时计划统计
	GET	`/api/opsflow/dashboard/template_stats/`	模板统计
Audit	GET	`/api/opsflow/audit/logs/`	审计日志
Log	GET	`/api/opsflow/logs/`	操作日志
Knowledge	CRUD	`/api/opsflow/knowledge/`	知识库
TemplateCategory	CRUD	`/api/opsflow/template-categories/`	模板分类
WebSocket	WS	`/ws/opsflow/execution/{id}/`	执行状态推送

5. 核心模块依赖图



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